

| Parameter                | Details                                                                        |
|--------------------------|--------------------------------------------------------------------------------|
| Dataset Title            | Polish Companies Bankruptcy Dataset (4year.arff)                               |
| Repository & DOI         | UCI Machine Learning Repository (DOI: 10.24432/C5F600)                         |
| Data Provider / Citation | Tomczak, S., Zieba, M., & Tomczak, J. M. (2016)                                |
| Data Source Database     | Emerging Markets Information Service (EMIS)                                    |
| Target Population        | Polish industrial and commercial firms                                         |
| Observation Unit         | Single firm financial statement (annual balance sheet / income statement)      |
| Time Horizon (4th Year)  | Financial ratios measured at Year 4 to predict bankruptcy status 2 years later |
| Historical Period        | Bankrupt firms: 2000-2012; Operating firms: 2007-2013                          |
| Sample Size (4year.arff) | N = 9,792 total observations (515 bankrupt, 9,277 operating)                   |
| Class Imbalance          | 5.26% bankrupt (1) vs. 94.74% operating (0)                                    |
| Missing Values           | Represented as '?' or NA in raw data; handled via complete-case analysis       |

| Renamed Variable | Original Code | Financial Definition / Formula                       | Data Type              | Role in Model          |
|------------------|---------------|------------------------------------------------------|------------------------|------------------------|
| bankrupt         | class         | 1 = Bankrupt within 2 years; 0 = Remaining operating | Binary Integer (0/1)   | Dependent Variable (Y) |
| profitability    | X1            | Net profit / total assets                            | Continuous (Ratio)     | Predictor (X1)         |
| leverage         | X2            | Total liabilities / total assets                     | Continuous (Ratio)     | Predictor (X2)         |
| working_capital  | X3            | Working capital / total assets                       | Continuous (Ratio)     | Predictor (X3)         |
| current_ratio    | X4            | Current assets / short-term liabilities              | Continuous (Ratio)     | Predictor (X4)         |
| asset_turnover   | X9            | Sales / total assets                                 | Continuous (Ratio)     | Predictor (X5)         |
| log_assets       | X29           | ln(Total assets) [Proxy for firm size]               | Continuous (Log Scale) | Predictor (X6)         |

| Variable Code | Category              | Financial Definition / Formula                                                                                      |
|---------------|-----------------------|---------------------------------------------------------------------------------------------------------------------|
| X1            | Profitability         | Net profit / total assets                                                                                           |
| X2            | Liquidity & Solvency  | Total liabilities / total assets                                                                                    |
| X3            | Liquidity & Solvency  | Working capital / total assets                                                                                      |
| X4            | Liquidity & Solvency  | Current assets / short-term liabilities                                                                             |
| X5            | Liquidity & Solvency  | [(cash + short-term securities + receivables - short-term liabilities) / (operating expenses - depreciation)] * 365 |
| X6            | Profitability         | Retained earnings / total assets                                                                                    |
| X7            | Profitability         | EBIT / total assets                                                                                                 |
| X8            | Liquidity & Solvency  | Book value of equity / total liabilities                                                                            |
| X9            | Turnover & Efficiency | Sales / total assets                                                                                                |

|     |                       |                                                                                   |
|-----|-----------------------|-----------------------------------------------------------------------------------|
| X10 | Turnover & Efficiency | Equity / total assets                                                             |
| X11 | Profitability         | (Gross profit + extraordinary items + financial expenses) / total assets          |
| X12 | Liquidity & Solvency  | Gross profit / short-term liabilities                                             |
| X13 | Turnover & Efficiency | (Gross profit + depreciation) / sales                                             |
| X14 | Profitability         | (Gross profit + interest) / total assets                                          |
| X15 | Turnover & Efficiency | (Total liabilities * 365) / (gross profit + depreciation)                         |
| X16 | Liquidity & Solvency  | (Gross profit + depreciation) / total liabilities                                 |
| X17 | Turnover & Efficiency | Total assets / total liabilities                                                  |
| X18 | Profitability         | Gross profit / total assets                                                       |
| X19 | Profitability         | Gross profit / sales                                                              |
| X20 | Turnover & Efficiency | (Inventory * 365) / sales                                                         |
| X21 | Growth & Size         | Sales (n) / sales (n-1)                                                           |
| X22 | Profitability         | Profit on operating activities / total assets                                     |
| X23 | Profitability         | Net profit / sales                                                                |
| X24 | Growth & Size         | Gross profit (in 3 years) / total assets                                          |
| X25 | Liquidity & Solvency  | (Equity - share capital) / total assets                                           |
| X26 | Turnover & Efficiency | (Net profit + depreciation) / total liabilities                                   |
| X27 | Growth & Size         | Profit on operating activities / financial expenses                               |
| X28 | Liquidity & Solvency  | Working capital / fixed assets                                                    |
| X29 | Growth & Size         | ln(Total assets)                                                                  |
| X30 | Turnover & Efficiency | (Total liabilities - cash) / sales                                                |
| X31 | Turnover & Efficiency | (Gross profit + interest) / sales                                                 |
| X32 | Turnover & Efficiency | (Short-term liabilities * 365) / cost of goods sold                               |
| X33 | Liquidity & Solvency  | Operating expenses / short-term liabilities                                       |
| X34 | Liquidity & Solvency  | Operating expenses / total liabilities                                            |
| X35 | Profitability         | Profit on sales / total assets                                                    |
| X36 | Liquidity & Solvency  | Total sales / total assets                                                        |
| X37 | Turnover & Efficiency | (Current assets - inventories) / long-term liabilities                            |
| X38 | Liquidity & Solvency  | Constant capital / total assets                                                   |
| X39 | Profitability         | Profit on sales / sales                                                           |
| X40 | Liquidity & Solvency  | (Current assets - inventory - receivables) / short-term liabilities [Quick Ratio] |
| X41 | Turnover & Efficiency | Total liabilities / ((profit on operating activities + depreciation) * 12 / 365)  |
| X42 | Profitability         | Profit on operating activities / sales                                            |
| X43 | Turnover & Efficiency | (Receivables * 365) / sales                                                       |

|     |                       |                                                                                                                                           |
|-----|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| X44 | Turnover & Efficiency | $(\text{Receivables} \times 365) / \text{cost of goods sold}$                                                                             |
| X45 | Growth & Size         | $\text{Net profit} / \text{inventory}$                                                                                                    |
| X46 | Liquidity & Solvency  | $(\text{Current assets} - \text{inventory}) / \text{short-term liabilities}$                                                              |
| X47 | Turnover & Efficiency | $(\text{Inventory} \times 365) / \text{cost of goods sold}$                                                                               |
| X48 | Profitability         | $\text{EBITDA} / \text{total assets}$                                                                                                     |
| X49 | Profitability         | $\text{EBITDA} / \text{sales}$                                                                                                            |
| X50 | Liquidity & Solvency  | $\text{Current assets} / \text{total liabilities}$                                                                                        |
| X51 | Liquidity & Solvency  | $\text{Short-term liabilities} / \text{total assets}$                                                                                     |
| X52 | Turnover & Efficiency | $(\text{Short-term liabilities} \times 365) / \text{cost of goods sold}$                                                                  |
| X53 | Turnover & Efficiency | $\text{Equity} / \text{fixed assets}$                                                                                                     |
| X54 | Liquidity & Solvency  | $\text{Constant capital} / \text{total assets}$                                                                                           |
| X55 | Growth & Size         | $\text{Working capital}$                                                                                                                  |
| X56 | Turnover & Efficiency | $(\text{Sales} - \text{cost of goods sold}) / \text{sales}$                                                                               |
| X57 | Growth & Size         | $(\text{Current assets} - \text{inventory} - \text{short-term liabilities}) / (\text{sales} - \text{gross profit} - \text{depreciation})$ |
| X58 | Liquidity & Solvency  | $\text{Total costs} / \text{total sales}$                                                                                                 |
| X59 | Liquidity & Solvency  | $\text{Long-term liabilities} / \text{equity}$                                                                                            |
| X60 | Growth & Size         | $\text{Sales} / \text{inventory}$                                                                                                         |
| X61 | Growth & Size         | $\text{Sales} / \text{receivables}$                                                                                                       |
| X62 | Turnover & Efficiency | $(\text{Short-term liabilities} \times 365) / \text{sales}$                                                                               |
| X63 | Turnover & Efficiency | $\text{Sales} / \text{short-term liabilities}$                                                                                            |
| X64 | Growth & Size         | $\text{Sales} / \text{fixed assets}$                                                                                                      |